



Hall Ticket No-

Code: S123528

Stanley College of Engineering and Technology for Women(A)**B.E.(CSE/EEE/ECE/IT/CME/AI&DS) II Semester (Backlog) Examinations May-June-2023****Mathematics-II**

Time: 3 hours

Max.Marks:60

PART-A**Note: Answer all questions (Compulsory)****5 X 2=10M**

- | | | | |
|---|--|-----|----|
| 1 | Write standard form of Bernoulli's differential equation of first order. | 2 M | L1 |
| 2 | Find the Particular Integral of $(D^2 + 4^2)y = e^{-4x}$ | 2 M | L1 |
| 3 | Define the Gamma function. | 2 M | L1 |
| 4 | State Cayley-Hamilton theorem. | 2 M | L1 |
| 5 | Find the Laplace Transform of t^3 | 2 M | L1 |

PART-B**Note: Answer all questions****5 X 10= 50M**

- | | | | |
|-----------|---|-----|----|
| 6 | a. Find the Orthogonal Trajectory of the family of curves $r^n = a^n \cos n\theta$, where θ is parameter. | 5M | L4 |
| | b. Solve $(3x^2 + 6xy^2)dx + (6x^2y + 4y^3)dy = 0$ | 5M | L3 |
| OR | | | |
| | c. Solve $(3y^2 + 4x) dx + 3xydy=0$ | 5M | L4 |
| | d. Solve $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$ | 5M | L3 |
| 7 | a. Solve $\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 6x = 0$, given $x(0) = 0, \frac{dx}{dt}(0) = 15$ | 5M | L3 |
| | b. Solve $(D^2 - 6D + 25)y = e^{2x}$. | 5M | L3 |
| OR | | | |
| | c. Solve by the method of variation of parameters, $\frac{d^2y}{dx^2} + 4y = \tan 2x$. | 10M | L3 |
| 8 | a. Show that $\int_a^b (x-a)^m (b-x)^n dx = (b-a)^{m+n-1} \cdot \beta(m+1, n+1)$ | 5M | L3 |
| | b. Show that $\int_0^\infty \sqrt{x} e^{-x^3} dx = \frac{\sqrt{\pi}}{3}$ | 5M | L3 |
| OR | | | |
| | c. State and Prove the relation between Beta and Gamma. | 5M | L3 |
| | d. Prove that $\beta\left(m, \frac{1}{2}\right) = 2^{2m-1} \beta(m, m)$ | 5M | L3 |

- 9 a. Reduce the quadratic form into canonical form by an orthogonal transformation. Also state the nature of quadratic form

10M L2

$$8x_1^2 + 7x_2^2 + 3x_3^2 - 12x_1x_2 - x_2x_3 + 4x_3x_1$$

OR

- c. Find the characteristic equation of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence compute A^{-1} .

5M L2

- d. Find the Eigen values of $\text{adj } A$ and $A^2 - 2A + I$, where $A = \begin{pmatrix} 2 & 3 & 4 \\ 0 & 4 & 2 \\ 0 & 0 & 3 \end{pmatrix}$

5M L2

- 10 a. Find the Laplace transform of $t^3 e^{-3t}$.

5M L3

- b. Find the inverse Laplace transform of the $\frac{1}{s(s^2+a^2)}$ by convolution theorem.

5M L3

OR

- c. Solve, using Laplace transform method, $y''' + 2y'' - y' - 2y = 0$ with the conditions $y(0) = y'(0) = 0$ & $y''(0) = 6$

10M L3



Hall Ticket No-

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Stanley College of Engineering and Technology for Women(A)

B.E (CSE/CME/AI&DS/IT) II Semester (Backlog)Examinations May-June-2023

Data Structures with C

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- 1 Explain how to find the performance of an algorithm.
- 2 Define the ADT for Deque.
- 3 Explain types of linked lists.
- 4 Describe the tree traversals with an example.
- 5 What is hashing?

2 M
2 M
2 M
2 M
2 M

PART-B

Note: Answer all questions

5 X 10= 50M

- 6 a. With the help of suitable example , explain the various operations used in datastructures.

10 M

OR

- c. Write a heap sort algorithm for sorting a list of integers in ascending order.

5M

- d. Explain how linear search works with an example.

5M

- 7 a. Write a C program to implement stack using singly linked list.

5M

- b. Write a C function for inserting an integer element in to a circular queue of integers.

5M

OR

- c. Write a program to concatenate single linked lists.

5M

- d. Give brief description about the storage using arrays.

5M

- 8 a. Define Double linked list. Write an algorithm to delete an element from the existing linked

5M

- b. Discuss about linked representation of Sparse Matrix.

5M

OR

- c. Write a C program to implement stack using arrays.

5M

- d. Explain the various operations on a Single Linked List.

5M

- 9 a. Construct the binary tree of the following data 25, 30, 10, 9, 62, 5, 18, 43, 53.

5M

- b. Explain the Graph traversals with an example.

5M

OR

- c. Explain various graph representations with an example and mention the pros and cons of each representation.

5M

- d. Explain Kruskal's algorithm for obtaining the Minimal Spanning Tree of a graph.

5M

- 10 a. Write an algorithm to delete an element from the binary search tree.

5M

- b. Construct the B-Tree of order 4 of the following data 20, 30, 40, 15, 18, 16, 50, 25, 9,17,10

5M

OR

- c. Explain the various operations on AVL tree.

5M

- d. Explain Collision Resolution Methods.

5M

Hall Ticket No-

Code: S123522

Stanley College of Engineering and Technology for Women(A)

B.E(CSE, CME, AI&DS) II Semester (Backlog) Examinations May-June-2023
Chemistry

Time:3hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- | | | | | |
|---|--|-----|-----|------|
| 1 | Recall any four specifications of portable water. | 2 M | CO1 | BTL1 |
| 2 | What are photovoltaic cells? | 2 M | CO2 | BTL1 |
| 3 | Define polymerization and functionality with an example. | 2 M | CO3 | BTL1 |
| 4 | What is (are) the main constituent (s) of CNG. | 2 M | CO4 | BTL1 |
| 5 | List any four applications of Composites. | 2 M | CO5 | BTL1 |

PART-B

Note: Answer all questions

5 X 10= 50M

- | | | | | |
|---|---|-------|-----|------|
| 6 | a. Describe the principle, procedure involved in the determination of total hardness of water by EDTA method. | 10 M | CO1 | BTL2 |
| | OR | | | |
| | c. Summarize the causes and effects of scales. | 5 M | CO1 | BTL2 |
| | d. Explain any 5 factors that affect rate of corrosion of metals. | 5 M | CO1 | BTL2 |
| 7 | a. If Zn(s) /Zn ²⁺ (0.1M) is coupled with Cu(s) /Cu ²⁺ (0.01M) ,
(i) write the cell notation, (ii) Calculate EMF of the cell at 25°C
(Given $E_{Zn^{2+}/Zn}^0 = -0.76V$, $E_{Cu^{2+}/Cu}^0 = 0.34V$) | 2+3 M | CO2 | BTL3 |
| | b. Illustrate construction and working of Methanol-O ₂ fuel cell. | 5M | CO2 | BTL3 |
| | OR | | | |
| | c. Illustrate the determination of pH of a solution using Quinhydrone Electrode? | 5 M | CO2 | BTL3 |
| | d. List primary components of Li-ion battery and explain working of Li-ion battery during discharge. | 2+3 M | CO2 | BTL3 |
| 8 | a. Draw Molecular Energy Level Diagram for O ₂ . | 5 M | CO3 | BTL3 |
| | b. Explain the preparation and applications of Buna-S. | 5 M | CO3 | BTL3 |

P.T.O

OR

- c. Draw neat Molecular Energy Level Diagram for NO. 5 M CO3 BTL2
- d. Explain the preparation, applications of Kevlar. 5 M CO3 BTL2
- 9 a. Calculate Gross calorific value of coal containing the following composition with the help of Dulong's formula.
C = 80 % , H = 5% , Oxygen = 4 % , N = 3% , S = 3% , and Ash = 5% 5 M CO4 BTL3
- b. Explain proximate analysis of coal. 5 M CO4 BTL2

OR

- c. Illustrate fractional distillation of petroleum with a neat diagram. 10 M CO4 BTL3
- 10 a. Explain any 10 principles of Green Chemistry with examples. 10 M CO5 BTL2

OR

- c. Explain transesterification of bio-diesel with an example. 5 M CO5 BTL2
- d. Differentiate carbon nanotubes and nano wires. 5 M CO5 BTL2





Hall Ticket No. -

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Stanley College of Engineering & Technology for Women (A)

B.E (CSE/CME/AI&DS) II Semester (Backlog) Examinations-May-June-2023

Basics Electrical & Electronic Circuits

Time: 03 Hours

Max. Marks-60

Part-A

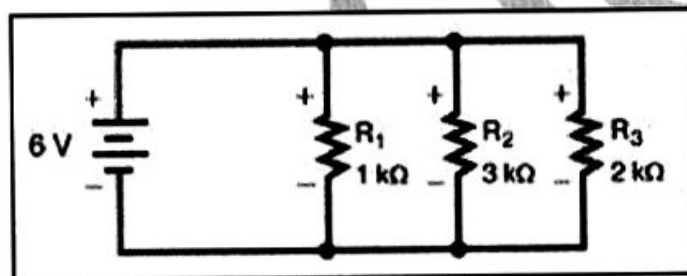
5X2=10M

1. State Kirchhoff's Laws. [2M BTL1]
2. Define Peak and RMS values. [2M BTL1]
3. Explain how a diode acts as a switch. [2M BTL2]
4. Discuss the applications of CB, CE & CC BJT amplifiers [2M BTL2]
5. List out the ideal characteristics of OPAMP. [2M BTL2]

Part-B

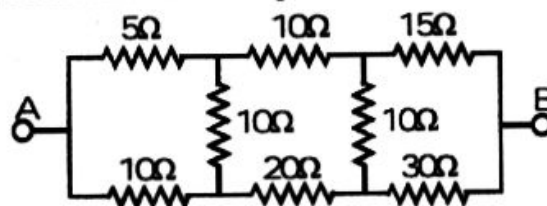
5X10=50M

6. a. State and prove Norton's theorem. [5M BTL2]
- b. Calculate the current through each resistor using Kirchhoff's laws. [5M BTL3]



OR

- c. State and prove Thevenin's Theorem. [5MBTL2]
- d. Find the effective resistance between points A and B [5M BTL3]



7. a) Show that $V_L = \sqrt{3} V_{ph}$ in 3-phase balanced Star connected system with the help of phasor diagram. [5M BTL2]
- b) Derive the Average and RMS value of current for sinusoidal waveform. [5M BTL3]

OR

- c) Explain series RL circuit when it is connected to $1 - \phi$ A.C supply. [5M BTL2]
d) Derive Average and RMS value of current for square wave form. [5M BTL2]
8. a) Draw and Explain Full wave rectifier circuit with wave forms. [6M BTL2]
b) Advantages and Disadvantages of Full wave rectifier circuit. [4M BTL1]
- OR
- c) Explain the working of half wave Rectifier with a neat circuit diagram along with necessary graphs and derive expression for its efficiency and ripple factor [5M BTL2]
d) Explain the working of CRO with neat block diagram. [5M BTL2]
9. a) Draw and explain the working of N-channel JFET. [6M BTL3]
b) Compare the performance of CE, CB and CC configurations. [4M BTL3]
- OR
- c) Explain the analysis of CE configuration. [5M BTL3]
d) Mention the advantages of JFET over BJT. [5M BTL3]
10. a) Discuss the different types of Negative Feedback. [5M BTL2]
b) Design the circuit of OP-AMP as Differentiator [5M BTL4]
- OR
- c) Explain the integrator and differentiator using op-amp. Derive its required equations [5M BTL-2]
d) Design the circuit of OP-AMP as Adder [5M BTL-3]

Stanley College of Engineering and Technology for Women(A)**B.E(CSE , CME , AI&DS) II Semester (Backlog) Examinations May-June-2023****Environmental Science**

Time: 3 hours

Max.Marks:60

PART-A**Note: Answer all questions (Compulsory)**

5 X 2=10M

- 1 Discuss the benefits of Dams. 2M BL2
- 2 Explain the ecological pyramid. 2M BL2
- 3 List out levels of Biodiversity. 2M BL1
- 4 Define Thermal pollution. 2M BL1
- 5 Explain the Basic principles of disaster mitigation. 2M BL2

PART-B**Note: Answer all questions**

5 X 10= 50M

- 6 a. Outline the environment and discuss the scope and importance of environmental studies. 5M BL4
- b. Explain deforestation and its effects on tribal people. 5M BL2

OR

- c. Illustrate environmental effect of mining. 5M BL4
- d. Summarize Various types of Resources. 5M BL2
- 7 a. Explain the Structure and function of the ecosystem. 5M BL2
- b. Outline various types of ecological succession. 5M BL4

OR

- c. Discuss various trophic levels in an ecosystem. 5M BL2
- d. Demonstrate Pond eco system. 5M BL3
- 8 a. Illustrate the various threats to Biodiversity. 5M BL4
- b. Discuss and explain the objectives and various strategies of biodiversity conservation. 5M BL2

OR

- c. Outline the Bio-geographical Zones of India. 5M BL4
- d. Explain levels and benefits of Biodiversity. 5M BL2
- 9 a. Explain the causes, effects, and control measures of Noise pollution. 5M BL2
- b. Describe salient features of the Water Act 1974. 5M BL2

OR

- c. Distinguish between primary and secondary pollutants and what is the relation between population and pollution. 5M BL2
- d. Out line air,water and soil pollutions and explain various approaches for controlling of all the mentioned pollutions with their merits and demerits. 5M BL2
- 10 a. Summarize climate change,global warming and acid rain. 5M BL5
- b. Illustrate the impact of the disaster on the environment, infrastructure, and development. 5M BL4

OR

- c. Classify the Disaster management cycle and disaster management in India. 5M BL4
- d. Outline Global warming and state various parameters responsible for the same. 5M BL2
